Experiment No. 3

Energy and Power Analysis of Signals

Experimental Procedure

1. Define the time vector and generate the selected signals.
2. Compute the squared magnitude of each signal.
3. Numerically integrate over a sufficiently large interval for energy.
4. Compute average power over an interval containing several periods for the periodic signal.
5. Compare numerical results with analytical expectations.
6. Classify the signals and record observations.

MATLAB Coding

```
% Experiment 3: Energy and Power Analysis
clc; clear; close all;

dt = 0.001;
t = 0:dt:5;

xE = exp(-t);
xP = sin(2*pi*5*t);

E_num = trapz(t,abs(xE).^2);

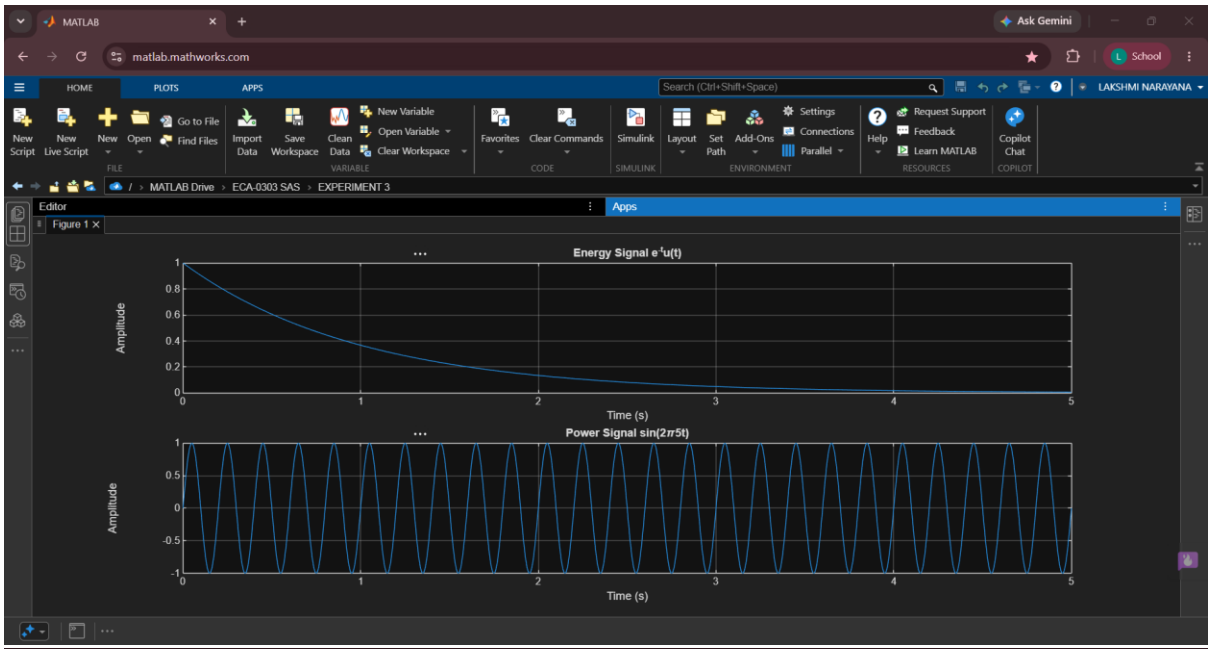
T0 = 1/5;
idx = t <= 2;
P_num = (1/(t(find(idx,1,'last'))-t(1))) * ...
      trapz(t(idx),abs(xP(idx)).^2);

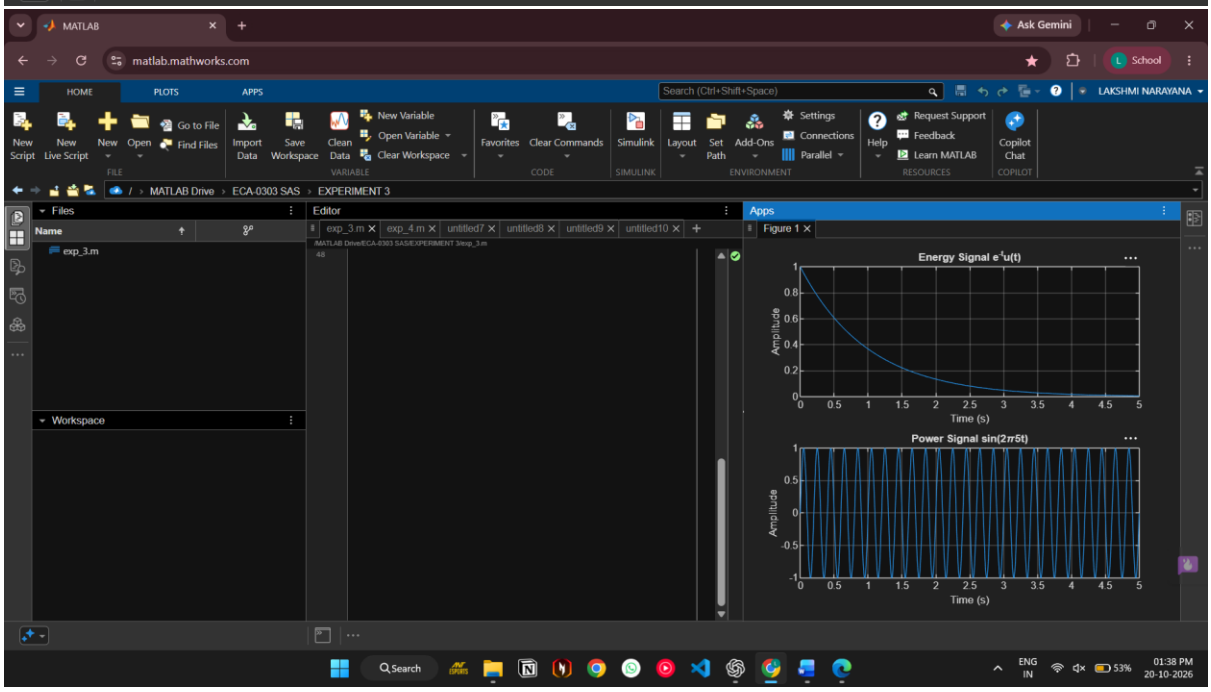
figure;
subplot(2,1,1);
plot(t,xE); grid on;
title('Energy Signal e^{-t}u(t)');
xlabel('Time (s)'); ylabel('Amplitude');

subplot(2,1,2);
plot(t,xP); grid on;
title('Power Signal sin(2pi5t)');
xlabel('Time (s)'); ylabel('Amplitude');

fprintf('Numerical energy = %g\n',E_num);
fprintf('Numerical average power = %g\n',P_num);
```

Output:


Energy Signal e-t u(t)
Amplitude
Time (s)
Power Signal sin(2πt)
Amplitude
Time (s)


Energy Signal e-t u(t)
Amplitude
Time (s)
Power Signal sin(2πt)
Amplitude
Time (s)